

CoderCommands Architecture

What changed to make this project self-contained, the system as it now runs, and what that move actually costs and gains.

scope: ManPage/ (compose project "codercommands") 2026-08-15

01 What changed

Two services were pulled fully inside this project. Nothing else in the stack changed shape.

DOCKER-MCP

- ▶ `ManPage/docker-mcp/` is new — a full copy of `backend/docker-mcp/docker-mcp`'s working tree (source only, no `.git`). It already includes the earlier `docker_host` `ssh-override`, made before this copy was taken.
- ▶ `docker-compose.yml`: the service's `build:` path changed from `../backend/docker-mcp/docker-mcp` to `./docker-mcp`.

POSTGRES-MCP

- ▶ `ManPage/postgres-mcp/` is new — not a copy of anything. A small custom FastMCP server (`asynpg`-backed) with four tools: `list_schemas`, `list_tables`, `describe_table`, `run_query`. Each takes an optional `connection_uri` override, the same per-call pattern as `docker-mcp`'s `docker_host`.
- ▶ `docker-compose.yml`: new `postgres-mcp` service added, defaulting to this project's own `postgres` via the existing `POSTGRES_USER` / `POSTGRES_PASSWORD` / `POSTGRES_DB` vars. Replaces the third-party `crystaldba/postgres-mcp` image this project used to reach in the separate `../POSTGRES` project.
- ▶ The `mcp-shared` external Docker network — previously required so `fastapi` could reach `postgres-mcp` across projects — is removed entirely.

- ▶ `api/main.py`: the `postgres-mcp` proxy route simplified to plain Streamable HTTP, same shape as `docker-mcp`'s. The old SSE-specific `/messages` side-channel route (needed only by the third-party server's transport) is gone.

TWO BUGS, CAUGHT BY ACTUALLY BUILDING AND RUNNING IT

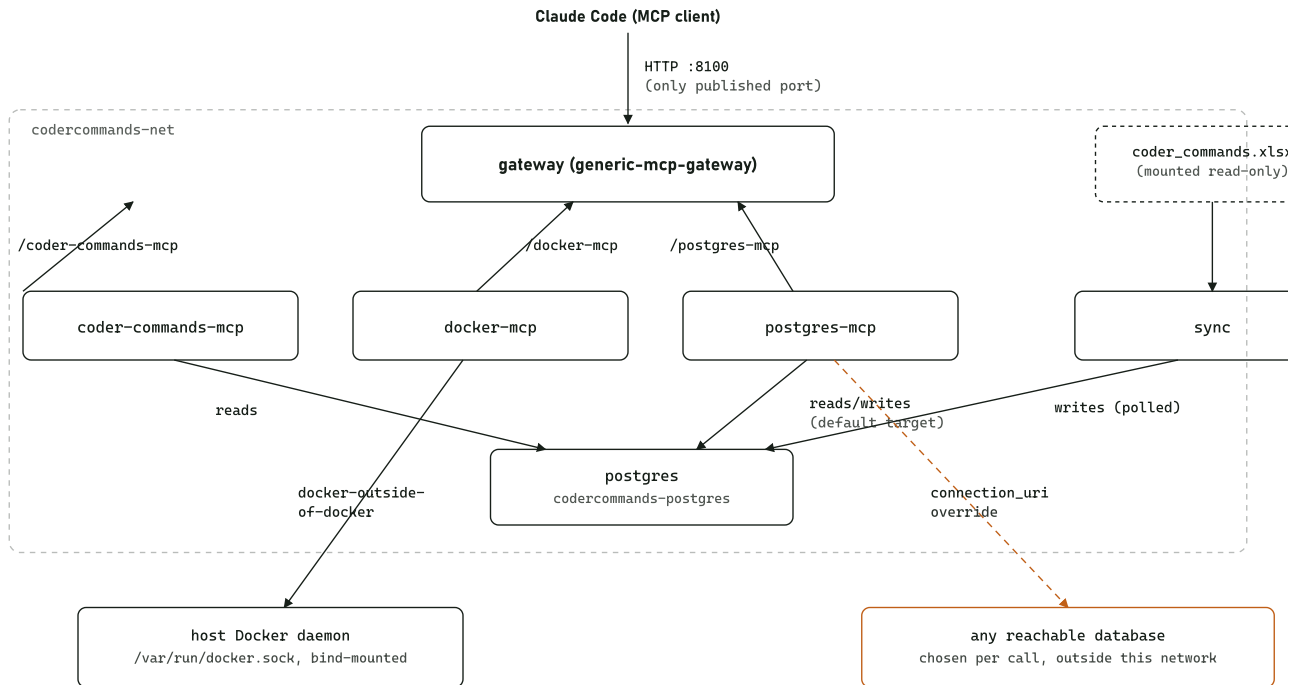
- ▶ `postgres-mcp/Dockerfile` copied `src/` *after* `pip install -e .` — a pattern carried over from `ignition-mcp`'s Dockerfile, likely broken there too but never exercised. Hatchling's editable install scans the source tree to auto-detect the package; installing before the copy fails metadata generation. Fixed the copy order.
- ▶ Its `HEALTHCHECK` used `curl -f` against `/mcp`, which 406s on a plain GET (Streamable HTTP needs an `Accept: application/json, text/event-stream` header) — `-f` turned that expected 406 into a permanent failure. Fixed the header and dropped `-f`; any response now proves the process is alive.

RENAMING — MCP-SERVER → CODER-COMMANDS-MCP, FASTAPI → GATEWAY

- ▶ `ManPage/mcp_server/` → `ManPage/coder-commands-mcp/`; compose service `mcp-server` → `coder-commands-mcp`; container name → `coder-commands-mcp`. It was never a generic server — naming it as if it might be one was the actual confusion.
- ▶ Compose service `fastapi` → `gateway`; container `codercommands-fastapi` → `codercommands-mcp-gateway`; `FASTAPI_HOST_PORT` → `GATEWAY_HOST_PORT`; FastAPI app title `"CoderCommands Gateway"` → `"MCP Gateway"` — it mounts whatever MCP servers this project has, not something built around coder-commands specifically.
- ▶ Gateway route `/mcp` → `/coder-commands-mcp`, matching the new service name.
- ▶ Third bug, same cause as the other two — verifying, not assuming: `coder-commands-mcp`'s own DNS-rebinding protection hardcoded the old `Host: mcp-server:9000` allow-list. After the rename, the gateway's proxied requests arrive with `Host: coder-commands-mcp:9000` and were rejected with `421 Misdirected Request` until the allow-list was updated to match.
- ▶ Container name `codercommands-mcp-gateway` → `generic-mcp-gateway`, one more pass at the same complaint: still not generic enough on a second look. `mcp-gateway` itself was ruled out — `backend/mcp-gateway` already runs a container by that exact name, and Docker container names must be unique per host.

02 Current architecture

Six containers, one published port. `gateway` is the only thing reachable from outside the compose network; everything else is addressed only by service name on `codercommands-net`.

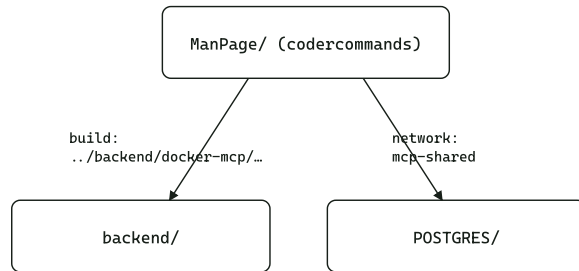


Every request enters through gateway's one published port and is routed by path prefix to one of three upstream MCP servers. `sync` is the only service with no proxied route — it only writes, on its own poll loop. `docker-mcp` is the only service that leaves the compose network, reaching the host's own Docker daemon. `postgres-mcp` is the only one with two destinations: its default target inside the network (solid), and any database reachable by a `connection_uri` override (dashed amber) — the same encoding used throughout this documentation series.

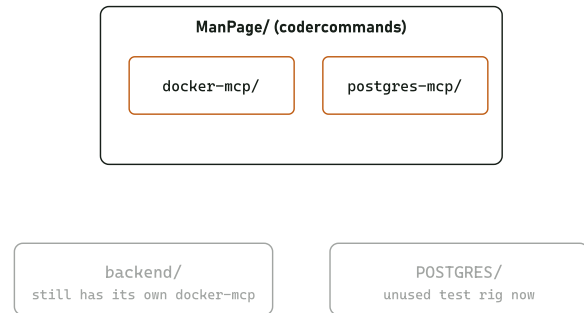
03 Folder structure, before → after

The dependency, not just the file listing, is what changed: `ManPage/` used to reach outside itself for two of its six services.

BEFORE - CROSS-PROJECT DEPENDENCIES



AFTER - SELF-CONTAINED



No arrows connect ManPage to backend/ or POSTGRES/ on the right — that absence is the change. docker-mcp/ and postgres-mcp/ became ManPage's own folders instead of paths reached at build or run time.

BEFORE

```

ManPage/
  docker-compose.yml
  api/
  mcp_server/
  sync/
  db/
  coder_commands.xlsx

../backend/docker-mcp/docker-mcp/
  ← built from here (build:)

../POSTGRES/compose.yaml
  ← reached over mcp-shared
  
```

AFTER

```

ManPage/
  docker-compose.yml
  docker-mcp/           ← new, own copy
  postgres-mcp/         ← new, own code
  api/
  coder-commands-mcp/   ← renamed from mcp_s
  sync/
  db/
  coder_commands.xlsx

../backend/docker-mcp/docker-mcp/
  (unchanged, no longer referenced)

../POSTGRES/compose.yaml
  (unchanged, no longer referenced)
  
```

04 What the move actually costs and gains

Gained: this project can now be copied to a cloud host on its own. Nothing in `ManPage/` needs `backend/` present at build time or `../POSTGRES` reachable at run time — both were real, working dependencies before this change, not hypothetical ones.

Cost — two copies of docker-mcp now exist. `backend/docker-mcp/docker-mcp` and `ManPage/docker-mcp` started identical but are independent from the moment of copying. A

fix made in one will not reach the other; keeping them in sync, if that's ever wanted, is a manual step from here on.

Cost — postgres-mcp lost capability, not just location. The third-party `crystaldba/postgres-mcp` this project used to run offered safe-SQL validation, `EXPLAIN` - plan analysis, index advice, and a restricted read-only mode. The replacement is a deliberately minimal four-tool server built to prove the override pattern, not a like-for-like port of every feature. Worth treating as a real tradeoff if any of that capability turns out to be needed.

Orphaned, not removed: `../POSTGRES` (the `postgres-mcp` + `pgadmin` test rig) still exists and still runs — it's simply no longer referenced by this project. Left as-is per instruction; nothing here decommissions it.

05 **Verified live**

Every claim above was exercised against the running stack, not just read out of the compose file.

CHECK	PATH	RESULT
postgres-mcp direct	<code>list_schemas,</code> <code>list_tables,</code> <code>run_query</code>	Real data — commands: 609 rows, sync_runs
postgres-mcp override	<code>connection_uri</code> → bogus host	Genuine connection attempt, not a silent fallback
gateway → postgres-mcp	<code>:8100/postgres-mcp</code>	Same result as direct call
gateway → docker-mcp	<code>:8100/docker-mcp</code>	list-containers returned the live container set
gateway → coder-commands-mcp	<code>:8100/coder-commands-mcp</code>	list_topics returned real topic counts
All six containers	<code>docker compose ps</code>	Up, healthchecked services report healthy
Post-rename regression pass	all three routes, rebuilt	421 from the stale Host allow-list caught and fixed, then all three re-verified

06 Open items

- ▶ backend/docker-mcp and ManPage/docker-mcp will drift — no sync mechanism exists between them today.
- ▶ ../POSTGRES is unused by this project but still running; decide whether to keep it around.
- ▶ postgres-mcp's four tools are a minimal set — revisit if crystalldb's safe-SQL validation or index advice turns out to matter.
- ▶ Neither docker-mcp nor postgres-mcp has been deployed to a cloud host yet — everything verified here is still local.

[ManPage/docker-compose.yml](#) · [ManPage/docker-mcp](#) · [ManPage/postgres-mcp](#) · [ManPage/coder-commands-mcp](#) · [ManPage/api/main.py](#)